Practical 2b: Student Record Management PWA

AIM : To create a Student Record Management Progressive Web Application with offline support and

installability.

Index.html

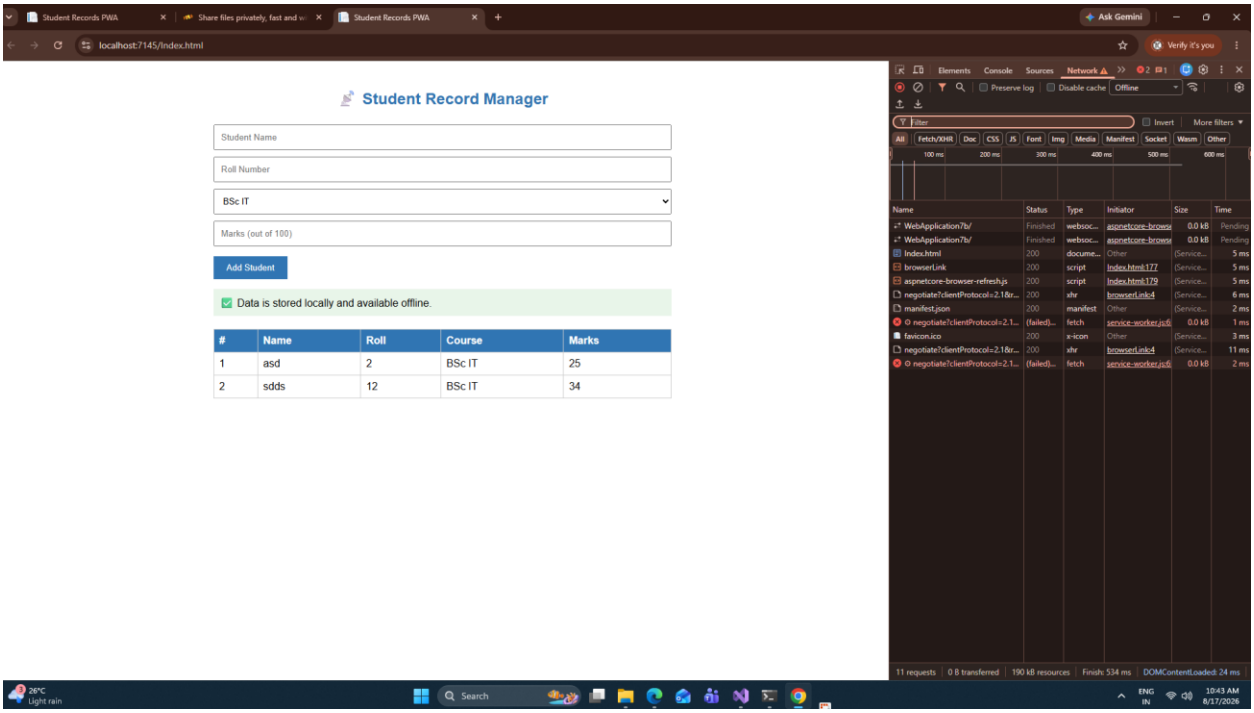
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Records PWA</title>
  <link rel="manifest" href="manifest.json" />
  <meta name="theme-color" content="#2E75B6" />
  <style>
    body {
      font-family: Arial, sans-serif;
      max-width: 600px;
      margin: auto;
      padding: 20px;
    }
    input, select {
      width: 100%;
      padding: 8px;
      margin: 5px 0;
      box-sizing: border-box;
    }
    button {
      background: #2E75B6;
```

```
      color: white;
      padding: 10px 20px;
      border: none;
      cursor: pointer;
    }
    table {
      width: 100%;
      border-collapse: collapse;
      margin-top: 20px;
    }
    th, td {
      border: 1px solid #ccc;
      padding: 8px;
      text-align: left;
    }
    th {
      background: #2E75B6;
      color: white;
    }
  </style>
</head>
<body>
  <h2> Student Record Manager</h2>
  <div>
    <input type="text" id="name" placeholder="Student Name" />
    <input type="text" id="roll" placeholder="Roll Number" />
    <select id="course">
      <option value="BSc IT">BSc IT</option>
      <option value="BSc CS">BSc CS</option
```

```
<option value="BCA">BCA</option>
    </select>
    <input type="number" id="marks" placeholder="Marks (out of 100)" />
    <button onclick="addStudent()">Add Student</button>
  </div>
  <table>
    <thead>
      <tr>
        <th>#</th>
        <th>Name</th>
        <th>Roll</th>
        <th>Course</th>
        <th>Marks</th>
      </tr>
    </thead>
    <tbody id="studentTable"></tbody>
  </table>
  <script>
    let students = JSON.parse(localStorage.getItem('students')) || [];
    function render() {
      const tbody = document.getElementById('studentTable');
      tbody.innerHTML = students.map((s, i) =>
        `<tr>
          <td>${i + 1}</td>
          <td>${s.name}</td>
          <td>${s.roll}</td>
          <td>${s.course}</td>
          <td>${s.marks}</td>
```

```
      </tr>`
    ).join('');
  }
  function addStudent() {
    students.push({
      name: document.getElementById('name').value,
      roll: document.getElementById('roll').value,
      course: document.getElementById('course').value,
      marks: document.getElementById('marks').value
    });
    localStorage.setItem('students', JSON.stringify(students));
    render();
  }
  render();
  if ('serviceWorker' in navigator) {
    navigator.serviceWorker.register('/sw.js');
  }
 </script>
</body>
</html>
```

Output:


Student Records PWA
Share files privately, fast and
Student Records PWA
Ask Gemini
localhost:7145/Index.html
Verify it's you
Student Record Manager
Student Name
Roll Number
BSc IT
Marks (out of 100)
Add Student
Data is stored locally and available offline.
Name Roll Course Marks
1 asd 2 BSc IT 25
2 sdds 12 BSc IT 34
Elements Console Sources Network
Preserve log Disable cache Offline
Filter Invert More filters
All Fetch/XHR Doc CSS JS Font Img Media Manifest Socket Wasm Other
100 ms 200 ms 300 ms 400 ms 500 ms 600 ms
Name Status Type Initiator Size Time
WebApplication7b/ Finished websoc... aspnetcore-browse 0.0 kB Pending
WebApplication7b/ Finished websoc... aspnetcore-browse 0.0 kB Pending
Index.html 200 docume... Other (Service... 5 ms
browserLink 200 script Index.html:177 (Service... 5 ms
aspnetcore-browser-refresh.js 200 script Index.html:179 (Service... 5 ms
negotiate?clientProtocol=2.1&r... 200 xhr browserLink:4 (Service... 6 ms
manifest.json 200 manifest Other (Service... 2 ms
negotiate?clientProtocol=2.1... (failed)... fetch service-worker.js:6 0.0 kB 1 ms
favicon.ico 200 x-icon Other (Service... 3 ms
negotiate?clientProtocol=2.1&r... 200 xhr browserLink:4 (Service... 11 ms
negotiate?clientProtocol=2.1... (failed)... fetch service-worker.js:6 0.0 kB 2 ms
11 requests 0 B transferred 190 kB resources Finish: 534 ms DOMContentLoaded: 24 ms
26°C Light rain
Search
ENG IN
10:43 AM 8/17/2026